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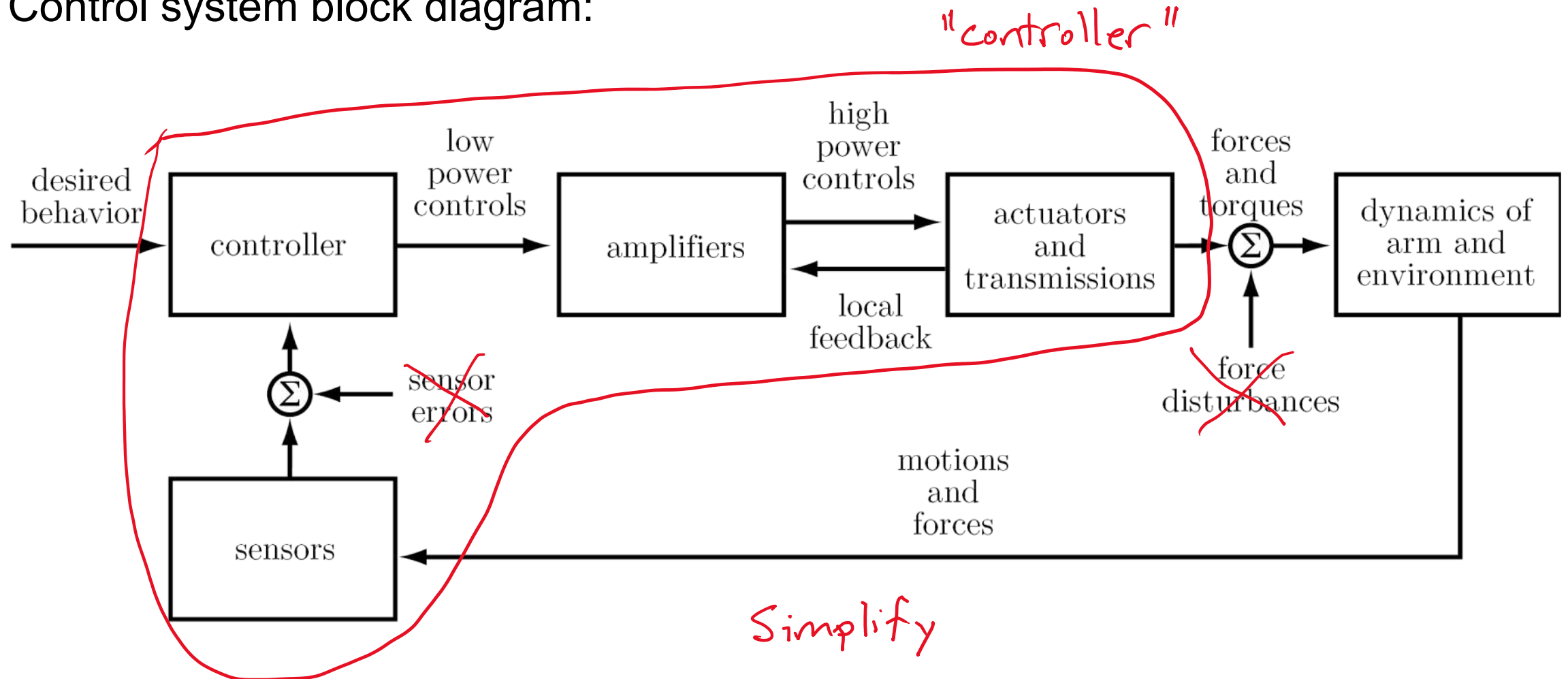
Important concepts, symbols, and equations

Example control objectives:

- **motion control**
- **force control**
- **hybrid motion-force control**
- **impedance control**

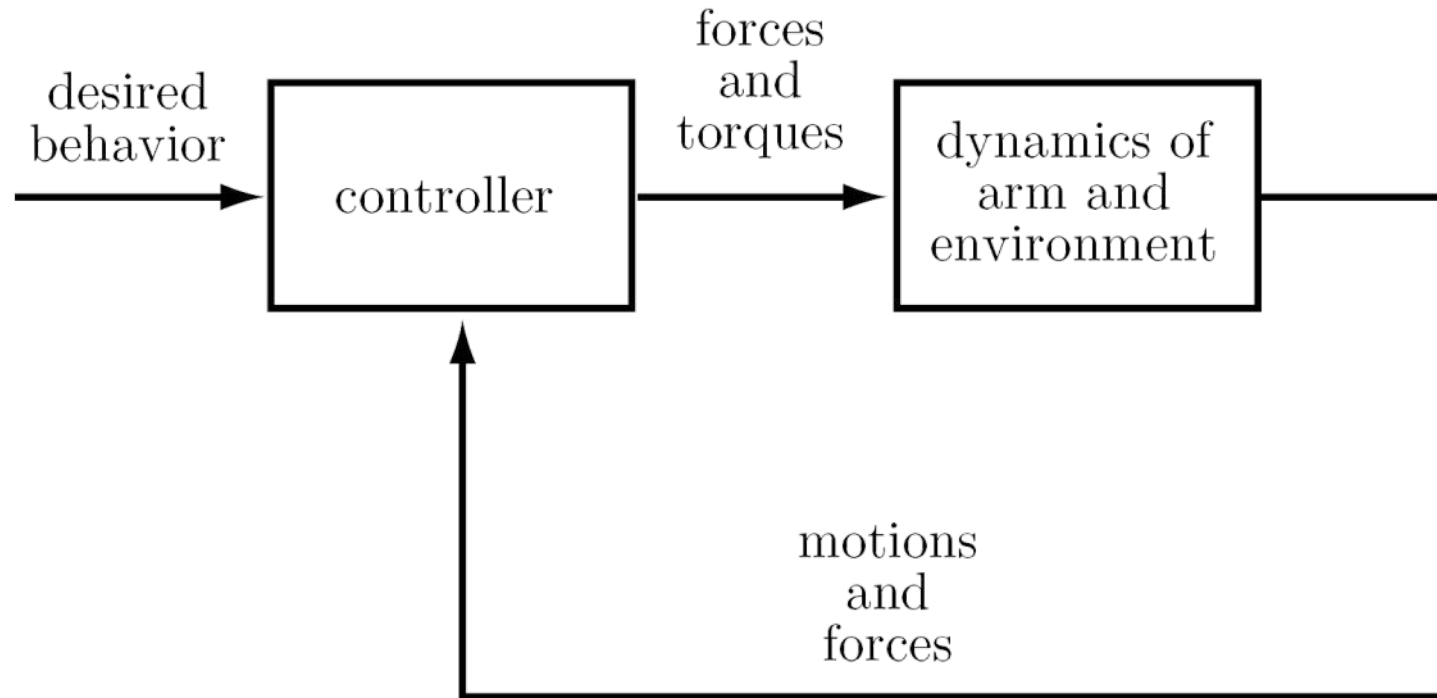
Important concepts, symbols, and equations (cont.)

Control system block diagram:



Important concepts, symbols, and equations (cont.)

Simplified block diagram:



Also assuming continuous-time (not discrete-time) control.

Important concepts, symbols, and equations (cont.)

For motion control,

reference: $\theta_d(t)$

actual: $\theta(t)$

error: $\theta_e(t) = \theta_d(t) - \theta(t)$

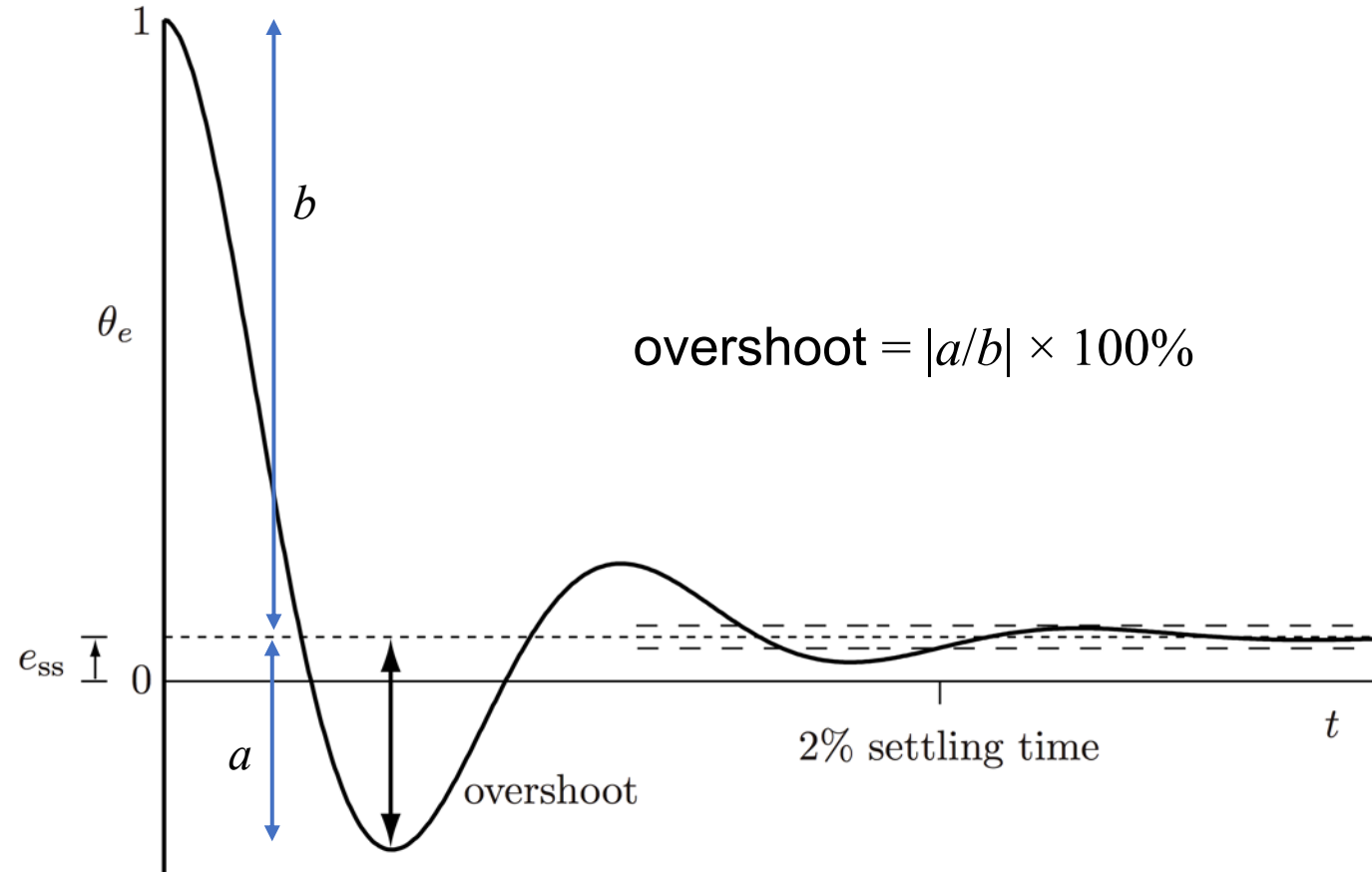
Unit step error response:

$\theta_e(t)$ starting from $\theta_e(0) = 1$

Steady-state error response: e_{ss}

Transient error response:

overshoot, settling time



Important concepts, symbols, and equations (cont.)

System dynamics, feedback controllers, and error response are often modeled by **linear ordinary differential equations**.

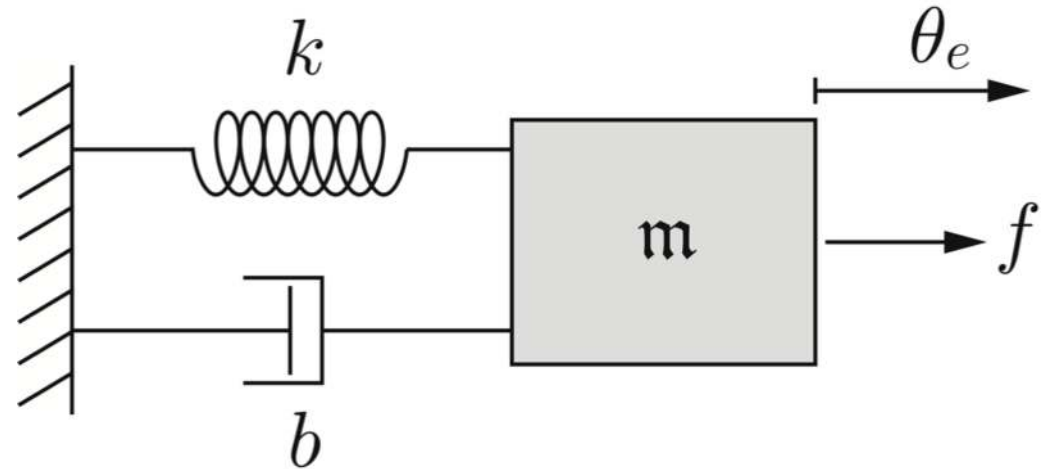
The simplest linear ODE exhibiting overshoot is second order, e.g.,

$$m\ddot{\theta}_e + b\dot{\theta}_e + k\theta_e = f$$

or, if $f=0$,

$$\ddot{\theta}_e + \frac{b}{m}\dot{\theta}_e + \frac{k}{m}\theta_e = 0$$

mbk model of error dynamics



k and b depend on the control law

Important concepts, symbols, and equations (cont.)

A more general p^{th} -order linear ODE:

$$a_p \theta_e^{(p)} + a_{p-1} \theta_e^{(p-1)} + \dots + a_2 \ddot{\theta}_e + a_1 \dot{\theta}_e + a_0 \theta_e = c \quad \text{nonhomogenous}$$

$$a_p \theta_e^{(p)} + a_{p-1} \theta_e^{(p-1)} + \dots + a_2 \ddot{\theta}_e + a_1 \dot{\theta}_e + a_0 \theta_e = 0 \quad \text{homogeneous}$$

$$\theta_e^{(p)} + a'_{p-1} \theta_e^{(p-1)} + \dots + a'_2 \ddot{\theta}_e + a'_1 \dot{\theta}_e + a'_0 \theta_e = 0$$

$$\theta_e^{(p)} = -a'_{p-1} \theta_e^{(p-1)} - \dots - a'_2 \ddot{\theta}_e - a'_1 \dot{\theta}_e - a'_0 \theta_e$$

Important concepts, symbols, and equations (cont.)

Defining a state vector $x = (x_1, x_2, \dots, x_p)$, you can write the p^{th} -order ODE as p first-order ODEs (a vector ODE).

$$x_1 = \theta_e,$$

$$x_2 = \dot{x}_1 = \dot{\theta}_e,$$

$$\vdots \quad \vdots$$

$$x_p = \dot{x}_{p-1} = \theta_e^{(p-1)}$$

$$\dot{x}(t) = Ax(t) \rightarrow x(t) = e^{At}x(0)$$

$$A = \begin{bmatrix} 0 & 1 & 0 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & 1 & 0 \\ 0 & 0 & 0 & \cdots & 0 & 1 \\ -a'_0 & -a'_1 & -a'_2 & \cdots & -a'_{p-2} & -a'_{p-1} \end{bmatrix} \in \mathbb{R}^{p \times p}$$

Important concepts, symbols, and equations (cont.)

$$\dot{x}(t) = Ax(t) \rightarrow x(t) = e^{At}x(0)$$

If $\text{Re}(s) < 0$ for all eigenvalues s of A , then the error dynamics are **stable** (the error decays to zero).

The eigenvalues are the roots of the **characteristic equation**

$$\det(sI - A) = s^p + a'_{p-1}s^{p-1} + \dots + a'_2s^2 + a'_1s + a'_0 = 0$$

Necessary conditions for stability: each $a'_i > 0$.

↳ the p^{th} -order ODE with s^k replacing $\Theta^{(k)}$ for $k=1 \dots p$.

These necessary conditions are also **sufficient** for first- and second-order systems.

Higher-order systems: use Routh-Hurwitz criterion.

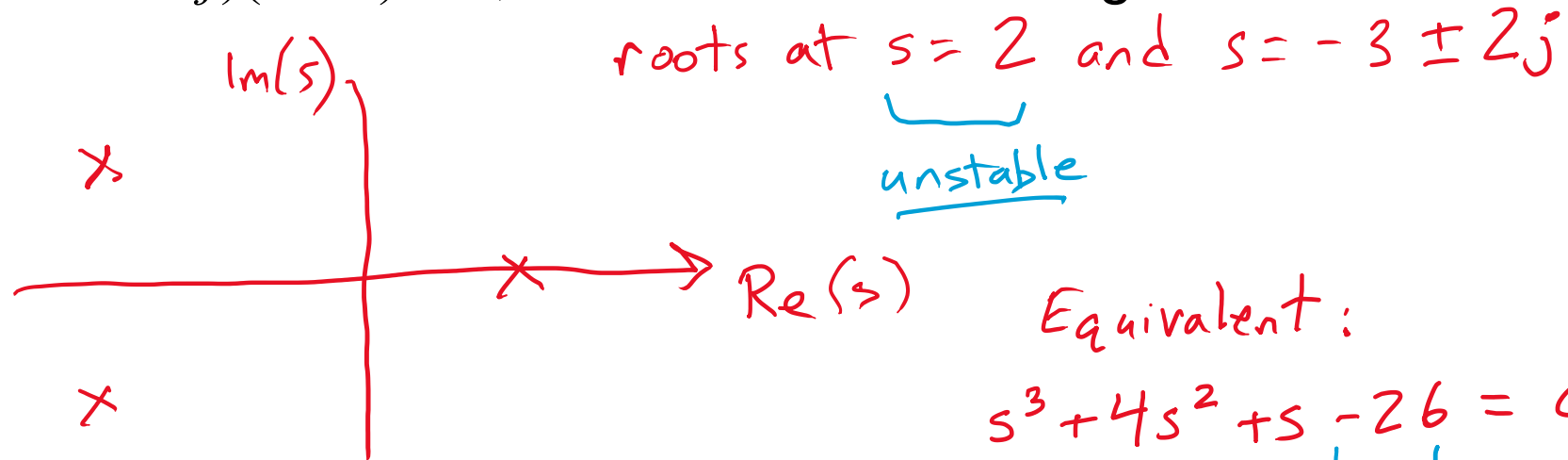
Types of control for the following tasks:

- Shaking hands with a human *impedance + "soft" motion control?*
- Erasing a whiteboard *hybrid*
- Spray painting *motion*
- Back massage *hybrid*
- Pushing an object across the floor with a mobile robot *motion*
- Opening a refrigerator door *hybrid*
- Inserting a peg in a hole *impedance (compliance) + motion*
- Polishing with a polishing wheel *force, maybe*
- Folding laundry *motion*

No single "correct" answer for most.

If the error dynamics characteristic equation is

$(s + 3 + 2j)(s + 3 - 2j)(s - 2) = 0$, does the error converge to zero?



Equivalent:

$$s^3 + 4s^2 + s - 26 = 0$$

does not satisfy
necessary condition

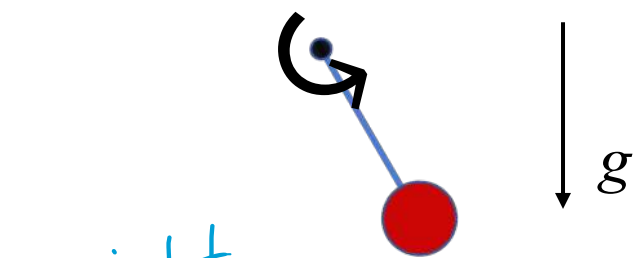
Note: if $x_1 = \text{error}$ and $x = (x_1, x_2, x_3)$, then $\dot{x} = Ax$, where

$$\begin{aligned}x_2 &= \dot{x}_1 \\x_3 &= \dot{x}_2 \\\dot{x}_3 &= 26x_1 - x_2 - 4x_3\end{aligned}$$

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 26 & -1 & -4 \end{bmatrix}$$

eigenvalues are $2, -3 \pm 2j$

You can choose a control law to be a virtual spring, a virtual damper, a virtual spring plus damper, or nothing. Which of these could stabilize an actuated pendulum with viscous friction to the upright configuration? To a horizontal configuration? To the downward configuration? Describe the transient and steady-state error response for each.



upright

Need P or PD
P has to overcome destabilizing "spring" of gravity.

spring : P control
damper : D control
both : PD

Natural viscous damping provides stabilization, but D term can allow better trans response.
 $e_{ss} = 0$

horizontal

Need P or PD. Will have $e_{ss} \neq 0$ if spring set point is exactly horizontal (due to gravity).

no controller is needed, but can improve trans response. $e_{ss} = 0$

downward